A Dual-Modal Visible and Near-Infrared Imaging System for Ocular Surface Analysis

Zhiwei Wang†, Meilu Zhu†, Yuxing Li*, Allie Lee, Edmund Y. Lam*, *Fellow, IEEE*

***Abstract*— Dry eye disease is a multifactorial ocular surface disorder characterized by tear film instability, hyperosmolarity, ocular surface inflammation, and neurosensory abnormalities. Meibomian gland dysfunction (MGD) is a leading cause of evaporative dry eye and is associated with quantitative and qualitative alterations of meibum. Clinically, MGD is commonly evaluated through meibomian gland (MG) morphology, including MG ratio, width, length, and tortuosity. Among these, the MG ratio, defined as the ratio of gland area to eyelid area, serves as an important quantitative biomarker for MG loss and disease severity. Accurate segmentation of the eyelid and MG regions is therefore essential for objective and reproducible MGD assessment. Dry eye disease is a multifactorial ocular surface disorder characterized by tear film instability, hyperosmolarity, ocular surface inflammation, and neurosensory abnormalities. Meibomian gland dysfunction (MGD) is a leading cause of evaporative dry eye and is associated with quantitative and qualitative alterations of meibum. Clinically, MGD is commonly evaluated through meibomian gland (MG) morphology, including MG ratio, width, length, and tortuosity. Among these, the MG ratio, defined as the ratio of gland area to eyelid area, serves as an important quantitative biomarker for MG loss and disease severity. Accurate segmentation of the eyelid and MG regions is therefore essential for objective and reproducible MGD assessment. The code will be released on our project page at `https://github.com/warren-wzw/DOSMG.git`.**

***Index Terms*— Ocular surface, Multimodal, Image fusion, Image segmentation, Meibomian gland.**

I. Introduction

Dry eye disease is a multifactorial ocular surface disorder characterized by tear film instability, hyperosmolarity, ocular surface inflammation, and neurosensory abnormalities [1]. Meibomian gland dysfunction (MGD) is a leading cause of evaporative dry eye and is associated with quantitative and qualitative alterations of meibum [2], [3]. Clinically, MGD is commonly evaluated through meibomian gland (MG) morphology, including MG ratio, width, length, and tortuosity [4], [5]. Among these, the MG ratio, defined as the ratio of gland area to eyelid area, serves as an important quantitative biomarker for MG loss and disease severity [6], [7]. Accurate segmentation of the eyelid and MG regions is therefore essential for objective and reproducible MGD assessment [8].

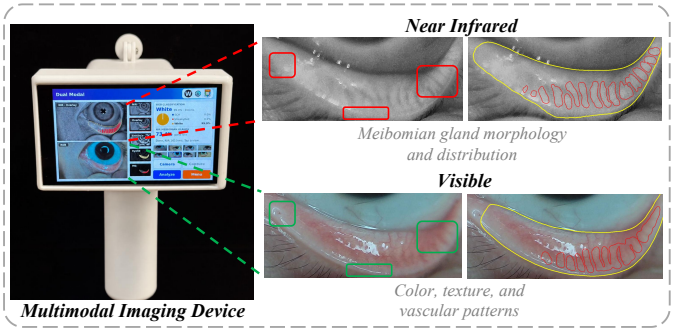


Fig. 1: Illustration of the proposed multimodal ocular imaging system. Near-infrared imaging highlights the morphology and distribution of meibomian glands, whereas RGB imaging provides rich appearance cues, including color, texture, and vascular patterns.

Non-contact infrared (NIR) meibography has emerged as the predominant modality for meibomian gland (MG) imaging, owing to its non-invasive nature, patient tolerability, and ability to provide high-contrast visualization of glandular structures [9]. Leveraging recent advances in artificial intelligence (AI), automated segmentation of the eyelid and MG regions from NIR images has been extensively explored, thereby circumventing the need for manual delineation, which is labor-intensive and subject to inter-observer variability [7], [10]. Nevertheless, NIR meibography alone suffers from intrinsic limitations. The eyelid margins are often poorly delineated, which compromises the accuracy of AI-based segmentation models and frequently results in erroneous MG ratio estimation [11]–[13]. In contrast, visible-light imaging provides complementary chromatic and textural information that is absent from NIR images. Specifically, the pinkish eyelid margin exhibits pronounced contrast against the sclera, offering salient cues for eyelid boundary delineation [14], [15]. However, visible-light images fail to resolve MG structures. *These complementary characteristics motivate the integration of multimodal visible and NIR imaging for ocular surface assessment and the diagnosis of MGD-related dry eye.* De-

This work is supported by the Innovation and Technology Fund (ITS/341/23) of Hong Kong, China.

Zhiwei Wang, Meilu Zhu, Yuxing Li, and Edmund Y. Lam are with the Department of Electrical and Computer Engineering, The University of Hong Kong, Hong Kong, China (e-mail: wangzw@eee.hku.hk; meiluzhu@hku.hk; yuxingli@hku.hk; elam@eee.hku.hk).

Zhiwei Wang is also with the College of Information Engineering, Zhejiang University of Technology, Hangzhou 310023, China (e-mail: wzw@zjut.edu.cn).

Allie Lee is with the Department of Ophthalmology, LKS Faculty of Medicine, The University of Hong Kong, Hong Kong, China (e-mail: aleeni@hku.hk).

†These authors contributed equally.

*Corresponding author(elam@eee.hku.hk).

spite these promising prospects, multimodal MGD diagnosis remains confronted with two non-trivial challenges [14], [16].

First, paired visible and NIR images must be spatially aligned and acquired within a sufficiently short temporal interval. Since the two modalities are captured sequentially rather than simultaneously, even a small acquisition gap can introduce eye motion, blinking, eyelid deformation, and subtle changes in gaze or illumination. Such factors may cause non-negligible spatial misalignment between the visible and NIR images [17], [18]. Reliable pixel-level correspondence is a prerequisite for cross-modal fusion; otherwise, misaligned features can propagate into the segmentation network [19] and degrade both eyelid and MG delineation, ultimately leading to inaccurate MG ratio estimation. In practice, however, most existing meibography devices are designed for single-modality imaging, typically NIR only, and therefore cannot provide spatially aligned visible–NIR image pairs [20]. This hardware limitation makes it impossible to directly exploit cross-modal complementarity. Consequently, developing a dedicated multimodal imaging device with a shared optical path and rapid modality switching is essential to acquire well-aligned data for reliable multimodal MGD diagnosis.

Second, effectively exploiting the complementary information of the two modalities is inherently difficult. Visible-light images provide strong chromatic and textural cues for eyelid boundary delineation, whereas NIR images reveal MG structures that are largely invisible under visible illumination [14]. However, simple concatenation, early fusion, or fixed weighted averaging treats the two modalities uniformly and may fail to dynamically emphasize the most reliable modality for different regions and tasks [21]. This difficulty is further compounded by the substantial domain discrepancies between the two modalities, which differ considerably in appearance, contrast, illumination, noise characteristics, and specular reflections. For instance, NIR meibography often contains strong reflections that obscure glandular texture, while visible-light images may suffer from uneven color, shadows, and ambiguous eyelid margins [21]–[23]. Meanwhile, MGs are thin, elongated, low-contrast, and highly variable in shape, length, and tortuosity, making joint eyelid and MG segmentation inherently unstable [7], [12], where even small boundary errors can substantially alter the measured MG area and MG ratio [10], [24]. Consequently, an effective fusion strategy must not only capture cross-modal correlations while suppressing modality-specific noise, redundancy, and ambiguity, but also remain robust to such domain discrepancies and segmentation ambiguity, especially when one modality is partially corrupted or uninformative.

To address these challenges, we develop a multimodal meibography device that employs a dual light source module and a dual-filter selector to enable RGB and NIR imaging through a shared optical path. The device is portable and acquires spatially aligned image pairs with only a small temporal gap. We further construct a multimodal dataset of paired visible and NIR ocular surface images with pixel-level annotations for the eyelid and MG regions. Building on this dataset, we propose a multimodal fusion model with dedicated RGB and NIR encoders and a cross-modal feature transfer and attention strategy for joint eyelid and MG segmentation. Experiments show that visible-light images are more suitable for eyelid segmentation, while dual-modality inputs are beneficial for MG segmentation. Simple fusion strategies fail to leverage the complementary information of the two modalities. In contrast, our proposed fusion strategy achieves the best MG segmentation performance and leads to more reliable MG ratio estimation. The main contributions of this work are summarized as follows:

1) We design a portable multimodal meibography device with a dual-light-source and dual-filter module, enabling spatially aligned visible and NIR imaging for ocular surface analysis.
2) We construct a multimodal MGD dataset with paired visible/NIR images and pixel-level eyelid and MG annotations.
3) We propose an AI-powered multimodal diagnosis system that integrates image enhancement, data annotation, multimodal segmentation, MG ratio calculation, and clinician interaction.
4) We develop a multimodal fusion model with RGB and NIR encoders and a tailored fusion strategy, and demonstrate its effectiveness for eyelid and MG segmentation.
5) Experimental results validate the complementary strengths of visible and NIR modalities, showing that visible light is more effective for eyelid segmentation, dual-modality inputs improve MG segmentation, and the proposed fusion strategy outperforms single-modality and early-fusion baselines.

These contributions collectively provide a practical and effective solution for multimodal dry eye and MGD diagnosis, and offer a foundation for further clinical translation of AI-assisted ocular surface analysis.

II. Related Work

A. Meibography Imaging Systems

Meibography visualizes the morphology of meibomian glands in vivo and has become a routine clinical examination for meibomian gland dysfunction (MGD) [9], [16]. Early contact meibography, relying on transillumination of the everted eyelid, often caused patient discomfort and constrained clinical adoption. Non-contact infrared meibography circumvented these limitations by capturing near-infrared (NIR) reflectance, subsequently becoming a staple in commercial ocular surface analyzers [9], [24]. To broaden accessibility, smartphone-based red-filter meibography has emerged as a cost-effective alternative [20], whereas optical coherence tomography (OCT) enables volumetric, three-dimensional visualization at the expense of heightened system complexity and cost [18]. Furthermore, while multispectral and hyperspectral approaches [15] as well as dual-mode systems [14] have been proposed to capture multi-spectral ocular responses, hardware fragmentation remains a persistent bottleneck: most existing platforms either operate in isolation or fail to deliver pixel-registered visible-NIR image pairs suitable for advanced fusion, while analysis largely relies on off-line workstations.In contrast, our work presents a compact, co-axial dual-modal

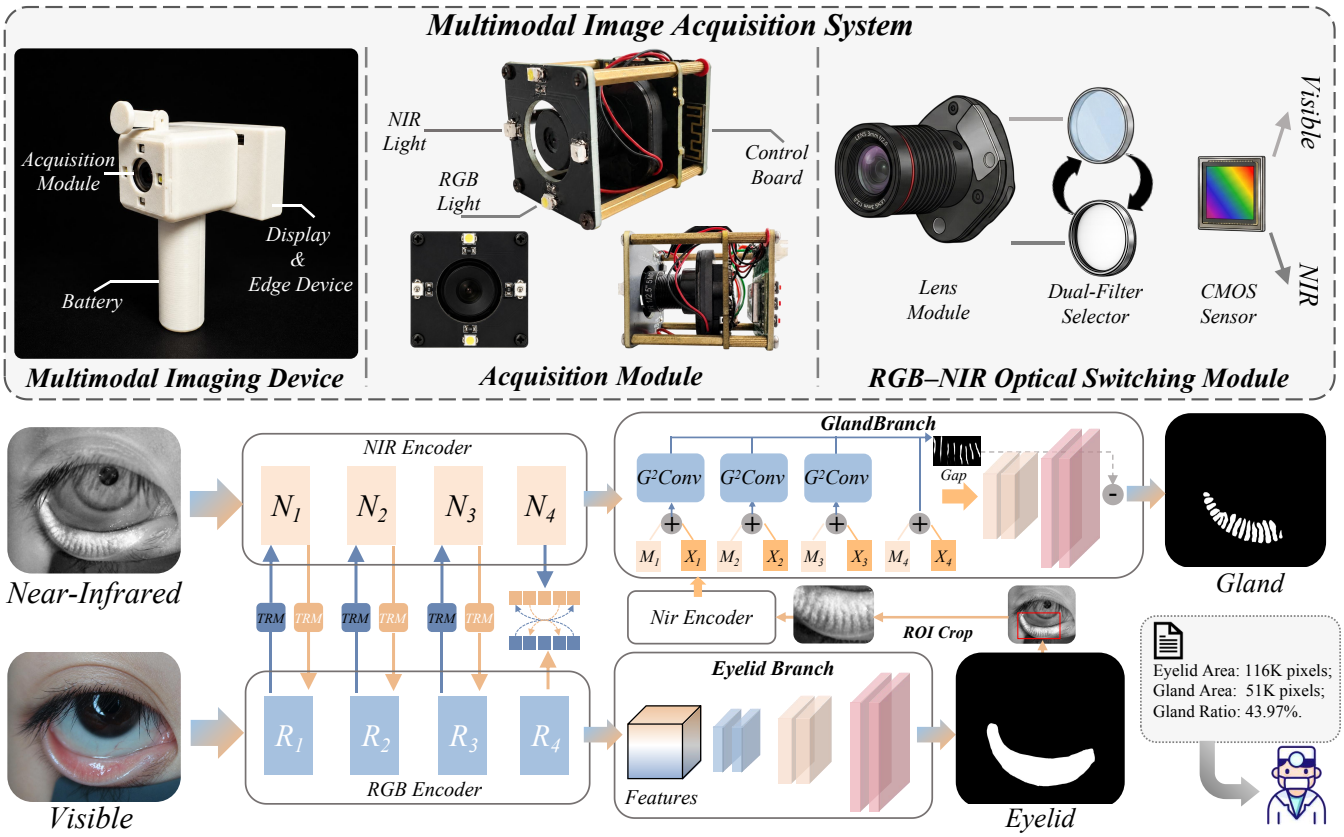


Fig. 2: Overall architecture of the proposed system. The system consists of a multimodal image acquisition module, an dual-modality representation learning module, and a dual-task segmentation network. Paired visible and near-infrared ocular surface images are captured and processed through asymmetric multimodal representation learning to exploit modality complementarity for joint eyelid and meibomian gland segmentation.

architecture that seamlessly acquires co-registered visible and NIR imagery through a shared optical path with minimal temporal intervals. By integrating edge-computed automated inference, this portable system overcomes current workflow silos, bridging the gap between multimodal acquisition and instantaneous, pixel-level meibomian gland assessment.

B. Eyelid and Meibomian Gland Segmentation

Manual delineation of the eyelid and meibomian gland (MG) regions is notoriously time-consuming and prone to substantial inter-observer variability, motivating the shift toward automated meibography analysis [24]. Early approaches relied on handcrafted pipelines—such as intensity thresholding, edge detection, and morphological operations—which remain acutely sensitive to uneven illumination, specular reflections, and low gland-to-background contrast. With the advent of deep learning, convolutional networks have substantially advanced segmentation accuracy. For instance, Setu et al. [23] employed a U-Net-based architecture for automatic MG segmentation and morphology assessment, while Saha et al. [7] quantified MG dropout by concurrently segmenting eyelid and gland regions. More recently, Zhu et al. [11] introduced a multi-task network featuring strip and boundary detection branches to better capture elongated gland structures; Paściak et al. [12] adopted an image-rotating Mask R-CNN for morphology evaluation; and Li et al. [10] developed a multicenter-validated AI framework for quantitative MG assessment.Despite these strides, current methodologies remain constrained by their exclusive reliance on NIR meibography. Because eyelid margins often suffer from poor anatomical delineation under NIR illumination, segmentation errors inevitably propagate and bias the denominator of the overall MG loss/dropout ratio [13]. Furthermore, densely distributed, low-contrast glands frequently manifest as structural adhesions and boundary fragmentations in predicted binary masks. To overcome these limitations, our work advocates a paradigm shift: we leverage visible-light imaging to secure robust, high-contrast eyelid delineation, coupled with an anatomy-guided gland branch designed to preserve fine-grained structural integrity during MG segmentation.

C. Multimodal Fusion for Segmentation

Multimodal fusion has been extensively studied to exploit complementary information across heterogeneous sensors. In natural scenes, RGB-thermal semantic segmentation networks—such as MFNet [25] and RTFNet [26]—typically adopt modality-specific encoders and fuse features via concatenation or element-wise summation. To model cross-modal interactions more adaptively, MMTM [27] recalibrates channel-wise features through squeeze-and-excitation-style transfer, while CMNeXt [28] supports arbitrary modality combinations by injecting auxiliary inputs into an RGB-centric

backbone. Furthermore, dynamic fusion strategies explicitly estimate modality reliability: DynMM [29] adaptively selects input-conditioned fusion paths, and PDF [30] predicts confidence-aware fusion weights backed by generalization guarantees.In medical imaging, multimodal fusion has been predominantly explored for multi-sequence MRI. For instance, mmFormer [31] combines modality-specific encoders with an inter-modal transformer to handle missing modalities, and M^{2}FTrans [32] introduces learnable fusion tokens with masked self-attention for incomplete brain tumor segmentation. More recently, Kim et al. [33] integrated local-window attention into U-Net for multimodal segmentation, and Pei et al. [34] proposed rebalanced fusion mechanisms to handle corrupted modalities. Despite these advancements, a fundamental limitation persists: existing methods either treat all modalities symmetrically or lock into a single primary modality for all targets, implicitly assuming that modal contributions are uniform across distinct segmentation objectives.In ocular surface imaging, however, this assumption fails: the eyelid margin is distinctively better delineated in visible-light spectra, whereas meibomian glands (MGs) are rendered almost exclusively under NIR illumination. This pronounced task-dependent modality preference breaks the symmetry of conventional fusion frameworks, motivating our asymmetric representation learning strategy—which constructs dedicated RGB-dominant and NIR-dominant pathways tailored specifically for eyelid and MG segmentation.

III. Multimodal Image Acquisition System

Existing meibography devices are primarily designed for clinical examination and morphological assessment, with certain limitations in portability, paired multimodal acquisition, and automated analysis. To address these limitations, we develop a portable ocular imaging system that integrates RGB–NIR imaging, edge computing, and result visualization, as shown in Figure 2. The system adopts a single-camera architecture with a shared optical path, equipped with a 1920 × 1080 CMOS sensor and a 6 mm lens with a 1/2.5-inch optical format and a 5 MP rating. Alternating RGB and NIR image acquisition is achieved by switching the illumination sources and optical filters. In the visible-light mode, a transparent filter preserves the color, texture, and vascular information of ocular tissues. In the NIR mode, an 850 nm illumination source and a matched narrow-band filter enhance the visibility of meibomian gland structures. The shared optical path reduces geometric discrepancies caused by different imaging viewpoints, while the acquisition interval between the two modalities is maintained at approximately 150 ms to minimize motion-induced spatial misalignment. Furthermore, the system incorporates an RK3566-based edge computing platform and a display module, enabling on-device image preprocessing, model inference, and segmentation visualization. This integration provides a unified solution for dual-modal imaging and automated eyelid and meibomian gland segmentation.

Based on the developed imaging system, we constructed the Dual-Modal Ocular Surface (DMOS) dataset by collecting RGB–NIR image pairs from 110 ocular samples. During acquisition, the device-to-eyelid position was kept stable to minimize spatial misalignment. For quality control, image pairs with noticeable misalignment were excluded through edge-overlay inspection. The remaining pairs underwent fine registration using a cross-modal dense matching algorithm [35], followed by manual screening to remove misaligned and near-duplicate acquisitions. This process yielded 510 valid RGB–NIR image pairs. Finally, pixel-level annotations of the eyelid and meibomian glands were manually created with assistance from a fine-tuned SAM 2 model [36], followed by manual verification and refinement to ensure anatomical accuracy and consistency. Representative RGB–NIR image pairs and segmentation masks are shown in Figure 3.

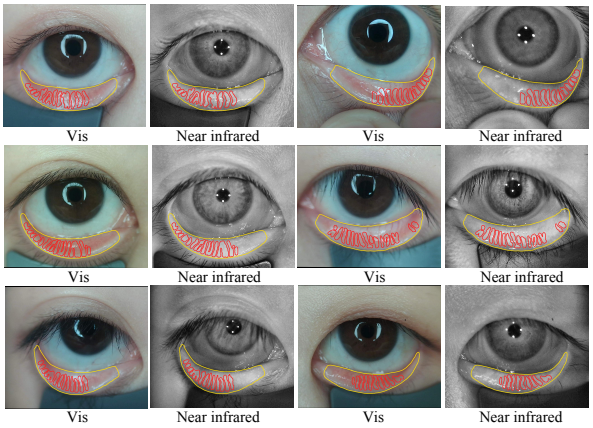


Fig. 3: Representative samples from the DMOS dataset. Each sample contains paired visible and near-infrared images with corresponding eyelid and meibomian gland segmentation masks.

IV. Anatomy-Aware Multimodal Segmentation

A. Overview

As illustrated in Figure 2, we propose a Dual-Modal Asymmetric Ocular Segmentation (DAOS) framework for joint segmentation of the eyelid and meibomian glands. Given a pair of spatially aligned RGB and NIR images, denoted as I_{vis} and I_{nir}, respectively, two independent encoders first extract multi-scale features to preserve complementary information from both modalities. Considering the distinct modality dependencies of the eyelid and meibomian glands, DAOS constructs task-specific representations dominated by RGB and NIR features, respectively, enabling task-oriented cross-modal information interaction.

Based on these representations, the network predicts the eyelid region and meibomian gland structures, where eyelid segmentation not only delineates anatomical boundaries but also provides spatial guidance for gland segmentation. Given the elongated and densely distributed morphology of meibomian glands, DAOS performs dual-scale prediction using both the full image and the eyelid region of interest (ROI), preserving global anatomical context while enhancing local structural details. Furthermore, the gland segmentation branch incorporates G^{2}Conv, which employs adaptive curved sampling along gland orientations to better capture elongated structures and improve gland continuity and separation. Finally, multi-task

and multi-scale supervision are jointly employed to optimize eyelid and meibomian gland segmentation.

B. Asymmetric Multimodal Representation Learning

The eyelid and meibomian glands exhibit distinct modality dependencies in RGB and NIR images. RGB images provide rich color, texture, and vascular information, facilitating eyelid boundary delineation, whereas NIR images better reveal the morphology and distribution of meibomian glands, serving as the primary information source for gland segmentation. Conventional symmetric multimodal fusion strategies typically employ uniform feature interaction mechanisms, overlooking the modality dependencies of different anatomical structures and potentially introducing redundant information and cross-modal interference. To address this limitation, we introduce Task-aware Residual Modulation (TRM), which enables task-specific asymmetric residual injection to preserve primary-modality representations while incorporating complementary information from the auxiliary modality. Specifically, I_{vis} and I_{nir} are fed into two independent encoders to extract multi-scale features R_l and N_l, respectively, defined as:

$$\begin{aligned} R_l &= E_{\text{vis}}^{(l)}(I_{\text{vis}}), \\ N_l &= E_{\text{nir}}^{(l)}(I_{\text{nir}}), \end{aligned} \tag{1}$$

Here, $l = 1, \ldots, L$ denotes the feature level.

For eyelid segmentation, RGB features provide sufficient anatomical boundary information. Therefore, lightweight linear residual modulation is adopted, where auxiliary NIR features are incorporated through a 1×1 convolution. For meibomian gland segmentation, NIR serves as the primary modality, and conditional nonlinear residual modulation is employed. Specifically, NIR and RGB features are first concatenated and projected through a 1×1 convolution, followed by a 3×3 depthwise convolution and GELU activation to extract local cross-modal features. Finally, an output projection generates residual features that are injected into the primary NIR representation. The fusion processes for the two tasks are formulated as follows:

$$\begin{aligned} F_l^{\text{EY}} &= R_l + \mathcal{P}_l^{\text{EY}}(N_l), \\ Z_l &= \phi\left(\mathcal{D}_l\left(\mathcal{P}_l^{\text{in}}([N_l; R_l])\right)\right), \\ F_l^{\text{MG}} &= N_l + \mathcal{P}_l^{\text{out}}(Z_l), \end{aligned} \tag{2}$$

where F_l^{EY} and F_l^{MG} denote the fused features for the two segmentation tasks, respectively. $\mathcal{P}_l^{\text{EY}}$, $\mathcal{P}_l^{\text{in}}$, and $\mathcal{P}_l^{\text{out}}$ are 1×1 convolutional projections, while $\mathcal{D}_l$ denotes a 3×3 depthwise convolution. ϕ represents the GELU activation function, and $[\cdot;\cdot]$ denotes channel-wise concatenation.

Both modulation strategies employ residual connections, with the convolutional weights of $\mathcal{P}_l^{\text{EY}}$ and $\mathcal{P}_l^{\text{out}}$ initialized to zero. This ensures that $F_l^{\text{EY}} = R_l$ and $F_l^{\text{MG}} = N_l$ at initialization, thereby preserving the original primary-modality representations. As training progresses, the residual branches gradually learn complementary information from the auxiliary modality. Through this asymmetric design, the eyelid task incorporates NIR information with minimal additional complexity, while the gland segmentation task leverages local NIR context to extract task-relevant complementary RGB features, yielding multimodal representations tailored to different anatomical structures.

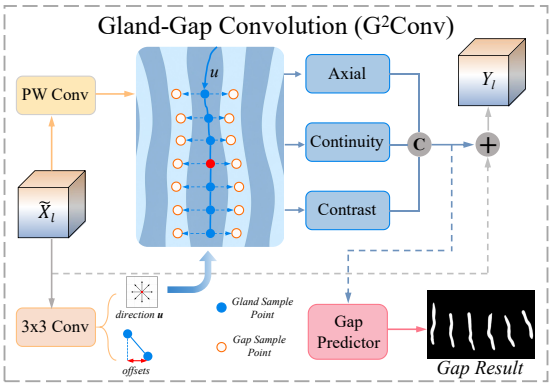


Fig. 4: Illustration of the G^{2}Conv. G^{2}Conv learns local gland orientations and lateral offsets to construct sampling paths, and integrates axial, continuity, and contrast cues to predict gland gaps, enhancing the structural representation of meibomian glands.

C. Anatomy-Guided Gland Segmentation Branch

Meibomian glands are confined to the eyelid and exhibit elongated, densely packed structures. Full-image segmentation may lose fine details due to downsampling, while local processing lacks global anatomical context. To address this, we introduce an eyelid-guided global–local feature fusion strategy. Specifically, an ROI box B^0 is derived from the eyelid mask through bounding-box extraction, margin expansion, and aspect-ratio adjustment. Paired RGB–NIR regions are then cropped from the original high-resolution images:

$$I_m^{\text{ROI}} = \mathcal{R}\left(\text{Crop}(I_m^0, B^0)\right), \quad m \in \{\text{vis}, \text{nir}\}, \tag{3}$$

where I_m^0 denotes the original image and $\mathcal{R}$ resizes the cropped region to $H \times W$. This preserves approximately $1.85\times$ higher effective spatial resolution than full-image input for typical samples. During training, ground-truth eyelid masks with random box perturbations are used, while inference relies solely on predicted masks.

To integrate global context with local details, we propose High-Resolution Gland Injection (HRGI). The ROI images are processed by the shared dual-modal encoders and asymmetric fusion pathway to obtain features X_l. Meanwhile, RoIAlign extracts spatially corresponding regions from full-image gland features M_l, which are injected through zero-initialized residual projections:

$$\widetilde{X}_l = X_l + \mathcal{Z}_l\left(\text{RoIAlign}(M_l, B)\right), \quad l = 1, \ldots, 4, \tag{4}$$

where B denotes the ROI coordinates mapped to the full-image input space, and $\mathcal{Z}_l$ is a zero-initialized 1×1 convolution. This preserves the local representation at initialization while progressively incorporating global context during training. The fused features are further processed by G^{2}Conv and the shared gland decoder. Both scales receive supervision during training, whereas only the ROI prediction is mapped back to the original image during inference.

Although HRGI integrates global context with local details, the fixed sampling grids of conventional convolutions remain inadequate for capturing the elongated, curved, and densely packed morphology of meibomian glands. Inspired by DSConv [37], we propose Gland-Gap Convolution (G^{2}Conv), as illustrated in Figure 4. G²Conv combines direction-adaptive curvilinear sampling, bilateral continuity modeling, cross-axis contrast enhancement, and gap prediction to jointly capture gland continuity and structural differences between adjacent glands. Specifically, given the HRGI-enhanced feature $\widetilde{X}_l$, a 1×1 convolution first reduces its channel dimensionality to obtain F_l. Meanwhile, a 3×3 convolution predicts a local direction, which is normalized into a unit vector $\mathbf{u}$, along with $2K$ lateral offset parameters, where $K = 4$. An adaptive curvilinear sampling path centered at position $\mathbf{p}$ is then constructed as follows:

$$\begin{aligned} \mathbf{p}_{\pm k} &= \mathbf{p} \pm k s_l \mathbf{u} + c_{\pm k}\mathbf{n}, \\ c_{\pm k} &= \sum_{j=1}^{k} m \tanh(o_{\pm j}), \quad k = 1, \ldots, K. \end{aligned} \tag{5}$$

where $\mathbf{n} = (-u_y, u_x)$ denotes the normal vector, $m = 1$ bounds the lateral displacement at each step, and $s_l = 8/r_l$ is the axial sampling step, with r_l denoting the downsampling factor at stage l. The direction is initialized vertically and the offsets are initialized to zero, yielding an initial straight sampling path aligned with the predominant gland orientation.

Let $A_k = F_l(\mathbf{p}_k)$ denote the features obtained through bilinear interpolation. G^{2}Conv first aggregates axial features along the curvilinear path and independently applies max pooling to the samples on either side of the center, producing Q^- and Q^+. A bilateral continuity response is then computed using a shared projection:

$$\begin{aligned} S_l &= \mathcal{W}_s([A_{-K}; \ldots; A_K]), \\ C_l &= \sigma(\mathcal{W}_g(Q^-)) \odot \sigma(\mathcal{W}_g(Q^+)) \odot \frac{Q^- + Q^+}{2}, \end{aligned} \tag{6}$$

where $\mathcal{W}_s$ denotes the axial feature aggregation operator, $\mathcal{W}_g$ is a shared 1×1 convolution, and $\odot$ represents element-wise multiplication. The bilateral response enhances weak regions supported by gland structures on both sides while reducing excessive extension beyond gland endpoints.

To further distinguish densely packed adjacent glands, we introduce cross-axis contrast modeling along the normal direction of the sampling path:

$$D_l = \mathcal{W}_c \left(\frac{1}{2K+1} \sum_{k=-K}^{K} \left[A_k - \frac{A_k^+ + A_k^-}{2} \right] \right), \tag{7}$$

where $A_k^{\pm} = F_l(\mathbf{p}_k \pm a_l \mathbf{n})$, $a_l = 12/r_l$ denotes the cross-axis sampling distance, and $\mathcal{W}_c$ is a learnable feature projection. By comparing responses along the gland path with those in its lateral neighborhoods, this mechanism enhances local contrast between glands and inter-gland gaps. The three feature components are then integrated through a zero-initialized residual projection:

$$Y_l = \widetilde{X}_l + \mathcal{W}_l^{\text{out}}([S_l; C_l; D_l]). \tag{8}$$

G^{2}Conv is independently applied to the first three ROI feature stages (S1–S3), while the fourth-stage features remain unchanged. Each G^{2}Conv also predicts a single-channel gap logit map from its input and the three aggregated feature components. The multi-scale gap logits are then combined and refined using the NIR ROI image:

$$\begin{aligned} G_l &= \mathcal{W}_l^{\text{gap}}([\widetilde{X}_l; S_l; C_l; D_l]), \\ \hat{G} &= \mathcal{D}\left(I_{\text{nir}}^{\text{ROI}}, G_1 + \text{Up}(G_2) + \text{Up}(G_3)\right), \end{aligned} \tag{9}$$

where $\mathcal{W}_l^{\text{gap}}$ denotes the gap prediction branch, Up aligns the gap logits to the S1 resolution, and $\mathcal{D}$ is a detail refinement module incorporating NIR image details.

D. Loss Function

To jointly optimize eyelid contours, meibomian gland structures, and inter-gland gaps, we formulate a multi-task, multi-scale loss function. The overall objective is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{EY}} + \mathcal{L}_{\text{MG}}^{\text{full}} + \mathcal{L}_{\text{MG}}^{\text{ROI}} + \lambda_{\text{gap}} \mathcal{L}_{\text{gap}}. \tag{10}$$

where $\mathcal{L}_{\text{EY}}$ denotes the eyelid segmentation loss, $\mathcal{L}_{\text{MG}}^{\text{full}}$ and $\mathcal{L}_{\text{MG}}^{\text{ROI}}$ represent the meibomian gland segmentation losses for the full-image and ROI branches, respectively, and $\mathcal{L}_{\text{gap}}$ is the inter-gland gap loss weighted by λ_{gap}.

For eyelid segmentation, cross-entropy (CE) and Dice losses are employed to optimize region prediction. For meibomian gland segmentation, we further introduce the clDice loss [38], which promotes structural continuity by measuring the consistency between predicted and ground-truth skeletons. The corresponding losses are defined as:

$$\begin{aligned} \mathcal{L}_{\text{EY}} &= \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Dice}}, \\ \mathcal{L}_{\text{MG}}^{s} &= \mathcal{L}_{\text{CE}}^{s} + \mathcal{L}_{\text{Dice}}^{s} + 0.5\mathcal{L}_{\text{clDice}}^{s}, \quad s \in \{\text{full}, \text{ROI}\}. \end{aligned} \tag{11}$$

Furthermore, to reduce erroneous connections between adjacent glands, multi-scale supervision is applied to the gap predictions of G^{2}Conv, jointly constraining the final and intermediate predictions:

$$\begin{aligned} \mathcal{L}_{\text{gap}} = \mathcal{L}_{\text{Dice}}(\hat{G}, Y_{\text{gap}}) \\ + 0.5 \sum_{l=1}^{3} \mathcal{L}_{\text{Dice}}(\text{Up}(G_l), Y_{\text{gap}}), \end{aligned} \tag{12}$$

where Y_{gap} denotes the ground-truth gap mask, $\hat{G}$ and G_l represent the final and intermediate gap predictions, respectively, and Up denotes resolution alignment.

V. Experiments

A. Implementation Details

All models are optimized using AdamW [39] with a weight decay of 1×10^{-4} for 100 epochs and a batch size of 4. The learning rate is initialized to 1×10^{-4} and gradually decayed to 1×10^{-6} using a cosine annealing schedule. The experiments are implemented in PyTorch 1.13.1 with CUDA 11.7 and performed on a single NVIDIA L40S GPU with 48 GB of memory under Rocky Linux 8.10.

B. Datasets and Evaluation Metrics

We evaluate segmentation performance on the DMOS and MGD1K [7]datasets. DMOS contains paired RGB–NIR images to assess dual-modality fusion, while MGD1K provides pixel-level annotations of eyelids and meibomian glands. For MGD1K, the same image is fed into both encoders to evaluate single-modality segmentation performance. DMOS is split into training and testing sets at a subject-wise ratio of 8:2. To enhance glandular textures, NIR images are sequentially processed using bilateral filtering [40], contrast-limited adaptive histogram equalization (CLAHE) [41], morphological top-hat transformation, and sharpening.

Four metrics are adopted to evaluate segmentation performance: eyelid Dice and MG Dice measure pixel-level segmentation accuracy; clDice assesses gland topology and continuity; and detF1 [42] evaluates the identification and separation of individual gland instances.

C. Segmentation Comparison and Analysis

We compare DOSMG with 11 representative segmentation methods, including single-modality methods such as SegFormer, Swin-Unet, and MedNeXt, as well as multimodal methods including MMTM, mmFormer, CMX, M2FTrans, DynMM, CMNeXt, PDF, and HLWA-UNet. The quantitative results are presented in Table I.

On the DMOS dataset, DOSMG achieves the best performance across all three meibomian gland segmentation metrics, with MG Dice, clDice, and detF1 scores of 77.650%, 85.277%, and 56.573%, respectively. Compared with the best-performing single-modality baseline, MedNeXt, DOSMG improves these metrics by 1.270, 1.527, and 2.976 percentage points, respectively. Compared with the strongest multimodal baseline, HLWA-UNet, the corresponding improvements reach 2.350, 2.354, and 5.196 percentage points. Notably, meibomian glands exhibit elongated and densely distributed structures, making segmentation quality dependent not only on pixel-level accuracy but also on structural continuity and the separation of adjacent glands. In particular, accurate gland segmentation requires preserving fine glandular structures while maintaining clear boundaries between neighboring glands. These characteristics make topology preservation and instance discrimination particularly important for reliable gland-level analysis. The consistent improvements in MG Dice, clDice, and detF1 demonstrate that DOSMG achieves more accurate gland segmentation while better preserving glandular structures and distinguishing individual gland instances. Moreover, the improvements across these complementary metrics indicate that the benefits extend beyond foreground region accuracy to the preservation of gland morphology and the identification of distinct gland instances. Furthermore, DOSMG achieves an eyelid Dice of 94.567%, demonstrating strong performance in both segmentation tasks. The standard deviations of 0.098 and 0.108 for MG Dice and clDice, respectively, further indicate consistent performance across repeated experiments.

On the MGD1K dataset [7], DOSMG also demonstrates excellent segmentation performance under single-modality input. As shown in Table I, DOSMG achieves the highest MG Dice and clDice scores of 83.150% and 91.540%, respectively, outperforming all competing methods on both metrics. In particular, clDice exceeds that of the second-best method, HLWA-UNet, by 0.273 percentage points, demonstrating the effectiveness of DOSMG in preserving glandular structural continuity. The improvements in both MG Dice and clDice indicate that DOSMG achieves accurate gland delineation while effectively preserving fine glandular morphology and topological structures. Notably, with the same image fed into both encoders, DOSMG achieves accurate gland segmentation while maintaining structural integrity. This further demonstrates the effectiveness of the proposed framework in capturing glandular structures without relying on complementary multimodal information. Furthermore, DOSMG achieves an eyelid Dice of 97.430% and a detF1 of 41.273%, demonstrating strong performance across both segmentation tasks. Fig. 5 presents a qualitative comparison of two representative samples from the DMOS dataset. Overall, all methods can effectively identify the eyelid region and recover its overall contour, while exhibiting noticeable differences in meibomian gland segmentation. The single-modality method SegFormer achieves relatively accurate eyelid segmentation but shows limited capability in recovering elongated glandular structures, with some glands appearing fragmented or missing. Multimodal methods leverage complementary features from RGB and NIR images to enhance segmentation performance. However, some methods still struggle to distinguish densely distributed adjacent glands, resulting in structural adhesion or the loss of local details.

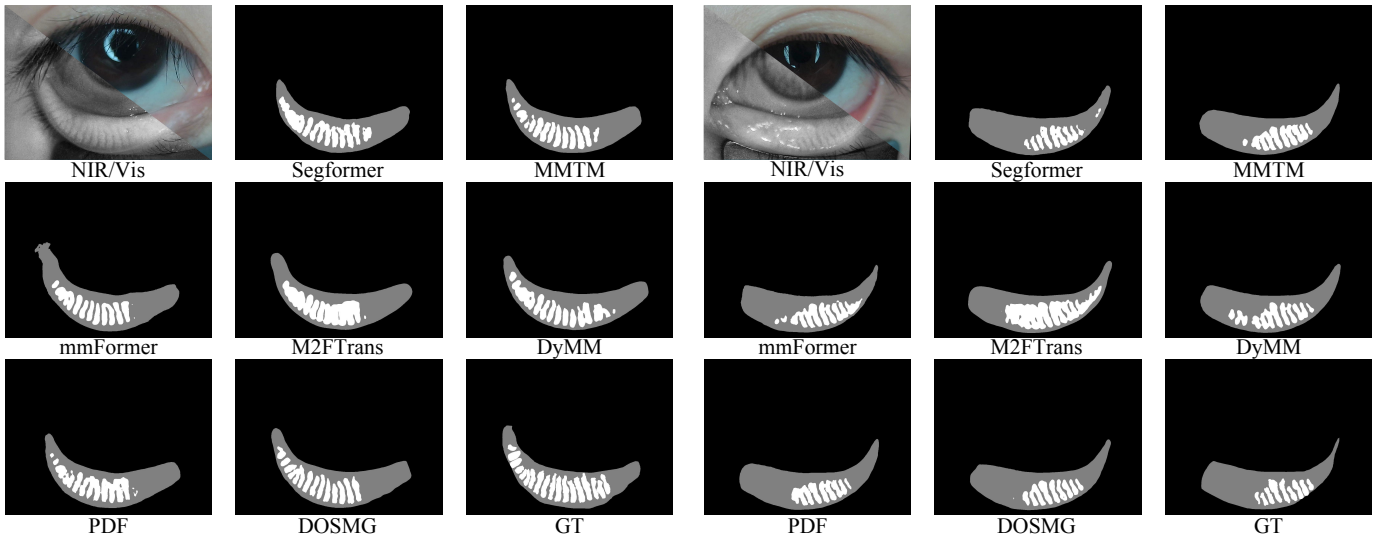
Specifically, in Sample 1, MMTM and mmFormer successfully identify the main glandular regions, but the morphology of some fine glands differs from the ground truth. M2FTrans and PDF exhibit noticeable adhesion between adjacent glands, affecting the accurate representation of individual gland structures. In contrast, DOSMG more accurately recovers the morphology of elongated glands while preserving the gaps between neighboring glands. In Sample 2, where the glands are more densely distributed, the differences in structural segmentation become more pronounced. SegFormer and MMTM can locate the main glandular regions but show incomplete recovery of some fine glands. M2FTrans exhibits substantial gland adhesion, while mmFormer and PDF produce incomplete glandular structures in certain regions. DOSMG, by comparison, preserves clearer gland boundaries and continuous elongated structures, producing predictions that more closely match the ground truth in terms of gland morphology and spatial distribution. Overall, DOSMG maintains accurate eyelid segmentation while better preserving elongated glandular structures and distinguishing adjacent glands, consistent with its quantitative advantages in MG Dice, clDice, and detF1.

D. Complexity Discussion

We further compare the model complexity and inference efficiency of different methods, as shown in Table Table II. Under a unified input resolution of 448×896, DOSMG requires 8.78M parameters and 59.72 GFLOPs, with an inference time of 14.06 ms per image. Compared with MedNeXt,

TABLE I: Quantitative comparison of DOSMG with state-of-the-art segmentation methods on the DMOS and MGD1K datasets.

Method	DMOS				MGD1K			
	Eyelid Dice ↑	MG Dice ↑	clDice ↑	detF1 ↑	Eyelid Dice ↑	MG Dice ↑	clDice ↑	detF1 ↑
SegFormer [43]	**94.597 ± 0.032**	72.810 ± 0.721	79.060 ± 0.530	16.227 ± 0.137	96.707 ± 0.032	81.680 ± 0.101	88.953 ± 0.116	29.053 ± 0.716
Swin-Unet [44]	92.037 ± 1.498	73.993 ± 0.366	77.240 ± 0.184	9.650 ± 0.310	97.000 ± 0.044	81.623 ± 0.025	86.867 ± 0.050	16.667 ± 0.289
MedNeXt [45]	94.137 ± 0.057	76.380 ± 0.326	83.750 ± 0.390	53.597 ± 1.253	97.383 ± 0.060	83.080 ± 0.020	91.073 ± 0.025	**41.427 ± 0.250**
MMTM [27]	94.320 ± 0.050	74.770 ± 0.151	81.423 ± 0.101	26.903 ± 1.040	97.410 ± 0.040	82.917 ± 0.085	90.527 ± 0.067	32.217 ± 0.348
mmFormer [31]	92.507 ± 0.475	70.097 ± 0.604	76.937 ± 0.581	24.863 ± 1.808	96.300 ± 0.106	81.460 ± 0.078	88.940 ± 0.026	26.693 ± 1.483
CMX [46]	94.383 ± 0.068	74.113 ± 0.652	80.463 ± 0.811	20.433 ± 3.675	97.343 ± 0.015	82.147 ± 0.100	89.807 ± 0.117	32.157 ± 1.844
M2FTrans [32]	93.103 ± 0.180	70.340 ± 1.355	77.100 ± 1.267	6.110 ± 2.066	97.060 ± 0.046	81.943 ± 0.104	89.420 ± 0.130	27.720 ± 0.062
DynMM [29]	93.467 ± 0.098	71.443 ± 1.116	78.320 ± 1.093	23.807 ± 0.760	97.423 ± 0.035	81.733 ± 0.155	89.773 ± 0.091	36.293 ± 1.168
CMNeXt [28]	94.557 ± 0.095	66.107 ± 0.666	71.457 ± 0.411	5.603 ± 2.792	**97.513 ± 0.067**	80.623 ± 0.087	88.447 ± 0.032	25.030 ± 1.478
PDF [30]	94.307 ± 0.081	75.217 ± 0.090	81.700 ± 0.122	25.480 ± 0.775	97.377 ± 0.038	82.963 ± 0.006	90.540 ± 0.061	32.230 ± 0.197
HLWA-UNet [33]	94.113 ± 0.055	75.300 ± 0.288	82.923 ± 0.257	51.377 ± 1.484	97.303 ± 0.035	83.067 ± 0.067	91.267 ± 0.080	40.633 ± 0.131
Ours	94.567 ± 0.050	**77.650 ± 0.098**	**85.277 ± 0.108**	**56.573 ± 1.133**	97.430 ± 0.053	**83.150 ± 0.040**	**91.540 ± 0.066**	41.273 ± 0.070



(a) Sample 1 (b) Sample 2

Fig. 5: Qualitative comparison of eyelid and meibomian gland segmentation results on two representative samples from the DMOS dataset. Gray and white regions indicate the eyelid and meibomian glands, respectively.

TABLE II: Comparison of model complexity and inference efficiency.

Method	Input Size	Params (M)	GFLOPs	Time (ms)
SegFormer [43]	448 × 896	7.43	25.81	6.47
Swin-Unet [44]	448 × 896	54.34	98.09	18.54
MedNeXt [45]	448 × 896	20.90	178.12	36.22
MMTM [27]	448 × 896	7.81	25.84	6.60
mmFormer [31]	448 × 896	19.53	45.03	10.98
M2FTrans [32]	448 × 896	7.00	32.84	16.96
DynMM [29]	448 × 896	59.90	75.05	5.85
CMX [46]	448 × 896	66.56	106.54	21.38
CMNeXt [28]	448 × 896	58.68	99.10	20.14
PDF [30]	448 × 896	8.48	52.18	7.97
HLWA-UNet [33]	448 × 896	15.40	80.08	6.13
Ours	448 × 896	8.78	59.72	14.06

DOSMG reduces the parameter count, computational cost, and inference time by 58.0%, 66.5%, and 61.2%, respectively, while achieving superior performance in MG Dice, clDice, and detF1. Compared with the multimodal method HLWA-UNet, DOSMG reduces the parameter count and computational cost by 43.0% and 25.4%, respectively, while further improving gland segmentation accuracy, structural continuity, and instance discrimination. These results demonstrate that DOSMG achieves a favorable balance between segmentation performance and computational efficiency.

E. Ablation Study

1) Ablation of Modality and Feature Fusion: To validate the complementarity of different modalities and the effectiveness of the proposed asymmetric feature fusion strategy, we evaluate the contributions of modality complementarity and asymmetric feature fusion. As shown in Table III, RGB only achieves an Eyelid Dice of 94.513%, outperforming NIR only at 93.160%, indicating that visible images are more effective for eyelid boundary delineation. In contrast, NIR only achieves an MG Dice of 77.127% and a detF1 of 55.113%, substantially outperforming RGB only, which achieves 65.227% and 16.573% on the corresponding metrics. These results demonstrate the complementary strengths of the two modalities for different segmentation tasks.

Building on this complementarity, the complete DOSMG achieves an Eyelid Dice of 94.567%, an MG Dice of 77.650%, and a detF1 of 56.573%, outperforming both single-modality variants. Removing TRM decreases MG Dice and detF1 by 0.443 and 1.080 percentage points, respectively, while removing GCMA leads to corresponding decreases of 0.320

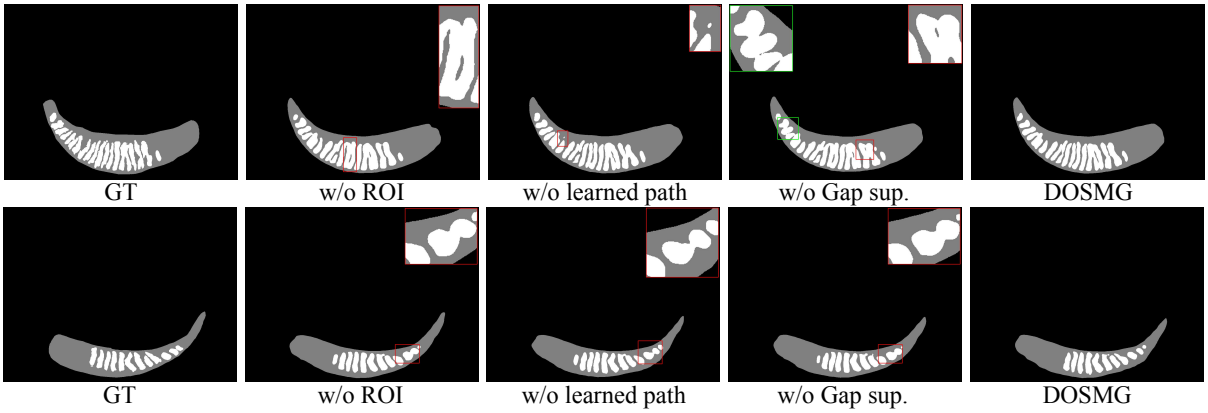


Fig. 6: Qualitative ablation comparison of DOSMG on two representative samples from the DMOS dataset.

and 1.216 percentage points. These results demonstrate the effectiveness of task-aware asymmetric feature fusion in exploiting modality-specific strengths and improving joint eyelid and meibomian gland segmentation.

2) Ablation of Anatomy-Guided Local Refinement: To evaluate the effectiveness of anatomy-guided local refinement, we investigate the contributions of ROI-based high-resolution processing, High-Resolution Gland Injection (HRGI), and clDice supervision. As shown in Table III, removing the ROI branch decreases MG Dice, clDice, and detF1 by 0.860, 0.934, and 4.486 percentage points, respectively, demonstrating the importance of high-resolution local features for accurate gland segmentation. As illustrated in Figure 6, removing the ROI branch leads to incomplete recovery of some fine gland structures, whereas the full model better preserves local gland morphology.

Furthermore, removing HRGI reduces MG Dice and detF1 by 0.527 and 3.106 percentage points, respectively, indicating that integrating global anatomical context with local high-resolution features improves gland segmentation accuracy and instance discrimination. Removing clDice supervision decreases clDice and detF1 by 0.717 and 4.470 percentage points, respectively, demonstrating the importance of topology-aware supervision for preserving gland continuity and separating adjacent glands. These results demonstrate that anatomy-guided local refinement, global–local feature interaction, and structural supervision jointly enhance the accuracy and structural integrity of meibomian gland segmentation.

3) Ablation of G^2Conv: To evaluate the effectiveness of G^2Conv in modeling meibomian gland structures, we first replace it with standard convolution and further investigate the contributions of its individual components. As shown in Table III, the complete model improves MG Dice, clDice, and detF1 by 0.663, 0.290, and 3.313 percentage points, respectively, compared with standard convolution, demonstrating the effectiveness of G^2Conv in capturing elongated and densely distributed glandular structures. Removing the adaptive sampling path decreases detF1 by 4.636 percentage points, indicating that adaptive sampling along local gland orientations facilitates gland morphology recovery and separation of adjacent glands. As illustrated in 6, removing the adaptive sampling path leads to local morphological deviations and erroneous connections between glands, whereas the complete model better preserves gland morphology and separation boundaries.

TABLE III: Ablation study of the proposed DOSMG framework. "w/o" denotes "without".

Method	Eyelid Dice ↑	MG Dice ↑	clDice ↑	detF1 ↑
NIR only	93.160 ± 0.190	77.127 ± 0.280	84.810 ± 0.390	55.113 ± 0.473
RGB only	94.513 ± 0.100	65.227 ± 0.796	74.173 ± 0.847	16.573 ± 1.181
w/o TRM	94.473 ± 0.166	77.207 ± 0.148	85.463 ± 0.201	55.493 ± 1.198
w/o GCMA	94.490 ± 0.156	77.330 ± 0.053	85.123 ± 0.265	55.357 ± 0.479
w/o clDice	94.473 ± 0.179	77.650 ± 0.095	84.560 ± 0.085	52.103 ± 1.021
w/o ROI	94.530 ± 0.302	76.790 ± 0.403	84.343 ± 0.548	52.087 ± 0.829
w/o HRGI	94.533 ± 0.025	77.123 ± 0.140	84.917 ± 0.506	53.467 ± 1.197
Standard Conv	94.520 ± 0.115	76.987 ± 0.242	84.987 ± 0.107	53.260 ± 1.507
w/o learned path	94.557 ± 0.248	77.353 ± 0.075	84.913 ± 0.185	51.937 ± 0.545
w/o axial	94.497 ± 0.038	77.363 ± 0.065	85.220 ± 0.050	55.957 ± 0.835
w/o continuity	94.470 ± 0.010	77.263 ± 0.097	85.277 ± 0.035	55.843 ± 1.080
w/o contrast	94.347 ± 0.032	77.327 ± 0.155	85.210 ± 0.269	53.023 ± 0.474
w/o Gap sup.	94.437 ± 0.055	$\mathbf{77.873 \pm 0.125}$	$\mathbf{85.473 \pm 0.133}$	54.487 ± 0.045
DOSMG (Ours)	$\mathbf{94.567 \pm 0.050}$	77.650 ± 0.098	85.277 ± 0.108	$\mathbf{56.573 \pm 1.133}$

Removing axial feature aggregation, bilateral continuity modeling, and cross-axis contrast modeling decreases MG Dice by 0.287, 0.387, and 0.323 percentage points, respectively. In particular, removing cross-axis contrast modeling reduces detF1 by 3.550 percentage points, highlighting its contribution to distinguishing adjacent glands. Furthermore, gap supervision improves detF1 from 54.487% to 56.573%, enhancing the separation of individual glands. The qualitative results in 6 further demonstrate that gap supervision helps reduce erroneous connections between adjacent glands, yielding predictions that more closely match the ground truth. Overall, G^2Conv enhances gland morphology recovery and instance discrimination through adaptive curvilinear sampling, structural feature modeling, and gap supervision.

VI. Conclusion

Acknowledgments

This work is supported by the Innovation and Technology Fund (ITS/341/23) of Hong Kong, China.

[1] J. P. Craig, K. K. Nichols, E. K. Akpek, B. Caffery, H. S. Dua, C.-K. Joo, Z. Liu, J. D. Nelson, J. J. Nichols, K. Tsubota *et al.*, "Tfos dews ii definition and classification report," *The ocular surface*, vol. 15, no. 3, pp. 276–283, 2017.

[2] B. Roczyński, I. Polok, O. Pniewska, and P. Polok, "Review meibomian gland dysfunction: Emerging diagnostic technologies and novel therapeutic approaches," *Journal of Education, Health and Sport*, vol. 95, pp. 73 540–73 540, 2026.

[3] J. Zhang and Z. Li, "From tarsal anatomy to tear-film homeostasis: A history-informed review of the meibomian gland in ocular surface biology," *Experimental Eye Research*, p. 111052, 2026.

[4] M. C. Lin, A. D. Graham, J. Ding, A. J. Fazioliu, D. K. Ma, and C.-H. Yeh, "Relationships between meibomian gland morphology and function," *Investigative Ophthalmology & Visual Science*, vol. 66, no. 8, pp. 1895–1895, 2025.

[5] Y. Deng, Q. Wang, Z. Luo, S. Li, B. Wang, J. Zhong, L. Peng, P. Xiao, and J. Yuan, "Quantitative analysis of morphological and functional features in meibography for meibomian gland dysfunction: diagnosis and grading," *EClinicalMedicine*, vol. 40, 2021.

[6] Y. Liu, Y. Ren, W. Li, W. Liu, and M. Ke, "A modified calculation formula for meibomian gland grading," *Experimental Eye Research*, vol. 250, p. 110166, 2025.

[7] R. K. Saha, A. M. Chowdhury, K.-S. Na, G. D. Hwang, Y. Eom, J. Kim, H.-G. Jeon, H. S. Hwang, and E. Chung, "Automated quantification of meibomian gland dropout in infrared meibography using deep learning," *The Ocular Surface*, vol. 26, pp. 283–294, 2022.

[8] A. Novo-Diez, M. Blanco-Vázquez, P. Arlanzón-Lope, C. Valencia-Sandonís, M. Calonge, M. J. González-García, and I. Fernández, "Diagnostic age-and sex-adjusted reference intervals of meibomian gland morphological features," *Contact Lens and Anterior Eye*, p. 102447, 2025.

[9] R. Arita, "Meibography: a japanese perspective," *Investigative ophthalmology & visual science*, vol. 59, no. 14, pp. DES48–DES55, 2018.

[10] L. Li, K. Xiao, T. Lai, K. Lai, J. Lin, Z. Ge, L. Liang, H. Huang, X. Zhang, L. Liu *et al.*, "Development and multicenter validation of an ai driven model for quantitative meibomian gland evaluation," *npj Digital Medicine*, vol. 8, no. 1, p. 403, 2025.

[11] W. Zhu, D. Liu, X. Zhuang, T. Gong, F. Shi, D. Xiang, T. Peng, X. Zhang, and X. Chen, "Strip and boundary detection multi-task learning network for segmentation of meibomian glands," *Medical Physics*, vol. 52, no. 3, pp. 1615–1628, 2025.

[12] A. Paściak, P. K. Piwowarczyk, D. R. Iskander, and D. H. Szczesna-Iskander, "Enhancing meibography based assessment of gland morphology by utilizing an image-rotating mask r-cnn approach," *Biomedical Signal Processing and Control*, vol. 109, p. 108045, 2025.

[13] N. B. Gokcinar and A. A. Karabulut, "Quantitative infrared meibography for ocular rosacea: lower eyelid meibomian gland loss as a diagnostic marker," *BMC ophthalmology*, vol. 26, no. 1, p. 4, 2025.

[14] Y. Li, P. W. Chiu, V. Tam, A. Lee, and E. Y. Lam, "Dual-mode imaging system for early detection and monitoring of ocular surface diseases," *IEEE Transactions on Biomedical Circuits and Systems*, vol. 18, no. 4, pp. 783–798, 2024.

[15] A. Zhang, "Multispectral and hyperspectral meibography," Oct. 25 2022, uS Patent 11,478,145.

[16] M. Robin, H. Liang, C. Baudouin, and A. Labbé, "In vivo meibomian gland imaging techniques: A review of the literature," *Journal Français d'Ophtalmologie*, vol. 43, no. 4, pp. e123–e131, 2020.

[17] S. L. Maskin and W. R. Testa, "Infrared video meibography of lower lid meibomian glands shows easily distorted glands: Implications for longitudinal assessment of atrophy or growth using lower lid meibography," *Cornea*, vol. 37, no. 10, pp. 1279–1286, 2018.

[18] H. S. Hwang, J. G. Shin, B. H. Lee, T. J. Eom, and C.-K. Joo, "In vivo 3d meibography of the human eyelid using real time imaging fourier-domain oct," *PLoS One*, vol. 8, no. 6, p. e67143, 2013.

[19] Y. Li, M. Tang, J. Hu, L. Li, and D. Liu, "Semantic–geometric dual alignment: A progressive co-optimization paradigm for misaligned multimodal medical image fusion," *Biomedical Signal Processing and Control*, vol. 120, p. 110043, 2026.

[20] G. H. Anissa, R. La Distia Nora, S. Widyawati, R. Sitompul, P. A. Yusuf, and A. Kekalih, "Red filter meibography by smartphones in patients with meibomian gland dysfunction: a validity and reliability study," *BMJ Open Ophthalmology*, vol. 9, no. 1, p. e001266, 2024.

[21] N. Song, C. Sun, Q. Li, S. He, J. Song, K. Yan, Y. Zhang, W. Wang, K. Cai, T. Fu *et al.*, "Parotid gland tumor computed tomography image segmentation and benign-malignant differentiation diagnosis," *Quantitative Imaging in Medicine and Surgery*, vol. 16, no. 9, p. 686, 2026.

[22] T. Hu, X. Nan, X. Zhou, Y. Shen, and Q. Zhou, "A dual-stream feature decomposition network with weight transformation for multi-modality image fusion," *Scientific reports*, vol. 15, no. 1, p. 7467, 2025.

[23] M. A. K. Setu, J. Horstmann, S. Schmidt, M. E. Stern, and P. Steven, "Deep learning-based automatic meibomian gland segmentation and morphology assessment in infrared meibography," *Scientific reports*, vol. 11, no. 1, p. 7649, 2021.

[24] K. Swiderska, M. L. Read, C. A. Blackie, C. Maldonado-Codina, and P. B. Morgan, "Latest developments in meibography: a review," *The ocular surface*, vol. 25, pp. 119–128, 2022.

[25] Q. Ha, K. Watanabe, T. Karasawa, Y. Ushiku, and T. Harada, "Mfnet: Towards real-time semantic segmentation for autonomous vehicles with multi-spectral scenes," in *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2017, pp. 5108–5115.

[26] Y. Sun, W. Zuo, and M. Liu, "Rtfnet: Rgb-thermal fusion network for semantic segmentation of urban scenes," *IEEE Robotics and Automation Letters*, vol. 4, no. 3, pp. 2576–2583, 2019.

[27] H. R. V. Joze, A. Shaban, M. L. Iuzzolino, and K. Koishida, "Mmtm: Multimodal transfer module for cnn fusion," in *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2020, pp. 13 286–13 296.

[28] J. Zhang, R. Liu, H. Shi, K. Yang, S. Reiß, K. Peng, H. Fu, K. Wang, and R. Stiefelhagen, "Delivering arbitrary-modal semantic segmentation," in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2023, pp. 1136–1147.

[29] Z. Xue and R. Marculescu, "Dynamic multimodal fusion," in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. IEEE, 2023, pp. 2575–2584.

[30] B. Cao, Y. Xia, Y. Ding, C. Zhang, and Q. Hu, "Predictive dynamic fusion," in *Proceedings of the 41st International Conference on Machine Learning*, vol. 235, 2024, pp. 5608–5628.

[31] Y. Zhang, N. He, J. Yang, Y. Li, D. Wei, Y. Huang, Y. Zhang, Z. He, and Y. Zheng, "mmformer: Multimodal medical transformer for incomplete multimodal learning of brain tumor segmentation," in *International conference on medical image computing and computer-assisted intervention*. Springer, 2022, pp. 107–117.

[32] J. Shi, L. Yu, Q. Cheng, X. Yang, K.-T. Cheng, and Z. Yan, "Mftrans: Modality-masked fusion transformer for incomplete multi-modality brain tumor segmentation," *IEEE journal of biomedical and health informatics*, vol. 28, no. 1, pp. 379–390, 2023.

[33] J. Kim, S. Jin, Y. Noh, H. Moon, M. Lee, and W. Noh, "Hybrid local-window-attention–assisted u-net model for multimodal medical-image segmentation," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2025, pp. 229–238.

[34] Y. Pei, X. Hu, Y. Zou, D. Hu, H. Chu, Y. Ma, S. Qiu, and G. Li, "When fusion fails: Corruption-aware rebalanced fusion for multi-modal medical image segmentation," *arXiv preprint arXiv:2609.10261*, 2026.

[35] Z. Wang, D. He, Y. Li, M. Zhu, and E. Y. Lam, "Geometry-aligned semantic matching for cross-modal planar image registration," *arXiv preprint arXiv:2610.03167*, 2026.

[36] N. Ravi, V. Gabeur, Y.-T. Hu, R. Hu, C. Ryali, T. Ma, H. Khedr, R. Rädle, C. Rolland, L. Gustafson *et al.*, "Sam 2: Segment anything in images and videos," in *International Conference on Learning Representations*, vol. 2025, 2025, pp. 28 085–28 128.

[37] Y. Qi, Y. He, X. Qi, Y. Zhang, and G. Yang, "Dynamic snake convolution based on topological geometric constraints for tubular structure segmentation," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, October 2023, pp. 6070–6079.

[38] S. Shit, J. C. Paetzold, A. Sekuboyina, I. Ezhov, A. Unger, A. Zhylka, J. P. Pluim, U. Bauer, and B. H. Menze, "cldice-a novel topology-preserving loss function for tubular structure segmentation," in *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2021, pp. 16 555–16 564.

[39] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations*, 2017.

[40] C. Tomasi and R. Manduchi, "Bilateral filtering for gray and color images," in *Sixth International Conference on Computer Vision (IEEE Cat. No.98CH36271)*, 1998, pp. 839–846.

[41] K. J. Zuiderveld, "Contrast limited adaptive histogram equalization," in *Graphics gems*, 1994.

[42] K. Sirinukunwattana, J. P. Pluim, H. Chen, X. Qi, P.-A. Heng, Y. B. Guo, L. Y. Wang, B. J. Matuszewski, E. Bruni, U. Sanchez *et al.*, "Gland segmentation in colon histology images: The glas challenge contest," *Medical image analysis*, vol. 35, pp. 489–502, 2017.

[43] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, "Segformer: Simple and efficient design for semantic segmentation with transformers," *Advances in neural information processing systems*, vol. 34, pp. 12 077–12 090, 2021.

[44] H. Cao, Y. Wang, J. Chen, D. Jiang, X. Zhang, Q. Tian, and M. Wang, "Swin-unet: Unet-like pure transformer for medical image segmenta-

tion," in *European conference on computer vision*. Springer, 2022, pp. 205–218.

[45] S. Roy, G. Koehler, C. Ulrich, M. Baumgartner, J. Petersen, F. Isensee, P. F. Jaeger, and K. H. Maier-Hein, "Mednext: transformer-driven scaling of convnets for medical image segmentation," in *International conference on medical image computing and computer-assisted intervention*. Springer, 2023, pp. 405–415.

[46] J. Zhang, H. Liu, K. Yang, X. Hu, R. Liu, and R. Stiefelhagen, "Cmx: Cross-modal fusion for rgb-x semantic segmentation with transformers," *IEEE Transactions on intelligent transportation systems*, vol. 24, no. 12, pp. 14 679–14 694, 2023.